Report: Programming Assignment – 1.5

1. Host Code Implementation

The host code reads matrix dimensions and initializes the input matrices with random data. Because Tensor Cores strictly require **fp16 inputs** but **output in fp32**, I implemented a helper function to convert the standard float matrices (A and B) into ***__half*** arrays before copying them to device memory. I allocated device memory, configured the grid and block dimensions, and used CUDA events to measure the execution times of both the PA1 baseline and the new Tensor Core kernels. Finally, the result is copied back to the host and verified against a reference output.

2. Tensor Core Kernel Implementation

The kernel uses the ***mma_m16n8k16_f16_f16_smem_row_col*** wrapper to perform warp-synchronous matrix multiplication using hardware Tensor Cores. Each 32-thread warp allocates shared memory for a 16x16 row-major A subtile, a 16x8 column-major B subtile, and a zero-initialized 16x8 C subtile. In a K/16 loop, threads cooperatively load inputs and synchronize via ***__syncwarp()*** before the MMA instruction, finally writing the accumulated output to global memory.

3. Performance Evaluation

To find the optimal performance, I ran experiments varying the number of warps per block on the maximum constraint problem size of 4096 * 4096 * 4096.

Configuration	Threads / Block	PA1 Baseline Time	Tensor Core Time	Speedup
4 Warps	128	327.407 ms	27.863 ms	11.75x
8 Warps	256	327.834 ms	27.556 ms	11.89x
16 Warps	512	327.383 ms	23.572 ms	13.88x
32 Warps	1024	327.422 ms	40.160 ms	8.15x

4. Conclusion

The Tensor Core kernel massively outperforms the PA1 scalar kernel across all valid configurations. I observed that the 32-warp configuration suffered a performance drop, likely due to shared memory resource exhaustion limiting SM occupancy. The 16-warp configuration (arranged logically in a 4*4 grid to compute a 64*32 output tile per block) provided the optimal balance of data reuse and latency hiding, achieving a peak **13.88x speedup**.

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